Asymmetric Capital Buffer Control

Manuscript skeleton, 10 October 2026

1 Introduction

The question is how an asymmetric penalty on a capital shortfall shapes a bank's recapitalisation policy and the volatility of its capital buffer. The model treats recapitalisation as impulse control under a fixed issuance cost and a regulatory cap on risky exposure.

Table 1: Headlines.

ID	Headline	Model rows	Scope note
K1	Under a binding regulatory cap, the ratio of the capital buffer's volatility deep in deficit to its volatility deep in surplus is fixed by the cap and carries no information about the asymmetry parameter λ_S.	M1, M2, M3	Holds wherever the optimal policy binds the cap near both ends of the state space, which is assumed and supported only by the numerical solve (Appendix B). Reading the discrete-time variance ratio as v_1^2/v_3^2 is a definition (Appendix D). Fails when $a_1 \geq a_3$ or when the cap does not bind (M3). The cap is that of Bayraktar et al. [2026], and the constant volatility of the log coordinate on the solvency branch is not claimed new.
K2	The asymmetry parameter and the fixed issuance cost are locally identified from the dividend barrier and the recapitalisation trigger, because they move the trigger in opposite directions.	M4, M5, M6, M7	Local identification from the thresholds, not an estimator. Rests on the hypotheses of Appendix B, among them the verification theorem, the envelope identities, the signs of $W'(x_L)$ and N', $\Psi(x_L) > 0$ and $\rho_L > \mu'(y^*)$. Fails if the two parameters moved the trigger in the same direction (M6). The fixed cost is charged per unit of liabilities (Appendix D). The responses of the barrier and the target to the fixed cost alone are not claimed new.

2 Model

Table 2: Model results.

ID	Claim	Statement	Lean	Proof
M1	The deficit-to-surplus variance ratio of the discrete-time recursion is λ_V^4, for every saturation scale.	$V_0 > 0$, $\theta > 0$, $\lambda_V > 0$, $V(z) = V_0 \exp\big(-2\log\lambda_V \tanh(z/\theta)\big) \Rightarrow$ $V(-z)/V(z) \to \lambda_V^4$ as $z \to \infty$.	`RateBased.Vresp_ratio_eq`, `RateBased.tendsto_tanh_atTop`, `RateBased.saturated_ratio_tendsto`, `RateBased.saturated_ratio_lambda4`	p. 7

ID	Claim	Statement	Lean	Proof
M2	The solvency branch of the cap binds exactly up to the crossing point $\bar{x}$, and there the z-volatility is the constant ν_1^2.	$0 < a_1 < a_3$, $a_2 < 1 \Rightarrow$ $\bar{x} - 1 = a_1(1-a_2)/(a_3 - a_1) > 0$, $u(x) = (x-1)/a_1$ for $x \le \bar{x}$, $u(x) = (x-a_2)/a_3$ for $x \ge \bar{x}$. If also $q > 0$, $1 < x \le \bar{x}$ and $\pi(x)\,x = u(x)$, then $\zeta(x) = \nu_1^2$.	`CapGeometry.hasDerivAt_zmap`, `CapGeometry.zvol2_eq`, `CapGeometry.xbar_sub_one`, `CapGeometry.one_lt_xbar`, `CapGeometry.solvency_binds`, `CapGeometry.liquidity_binds`, `CapGeometry.zvol2_solvency`	p. 7
M3	For any exposure policy that binds the cap near an end of the state space, the z-volatility tends to ν_1^2 at the distress boundary and to ν_3^2 deep in surplus.	$q > 0$, $0 < a_1 < a_3$, $a_2 < 1$. If $\pi(x)\,x = u(x)$ for all x near 1^+ then $\zeta(x) \to \nu_1^2$ as $x \to 1^+$. If $\pi(x)\,x = u(x)$ for all large x then $\zeta(x) \to \nu_3^2$ as $x \to \infty$. Both fail without $a_1 < a_3$ or without binding.	`CapGeometry.deficit_limit`, `CapGeometry.surplus_limit`, `CapGeometry.variance_ratio_is_cap_geometry`, `CapGeometry.control_surplus_limit_needs_a1_lt_a3`, `CapGeometry.control_constancy_needs_binding`	p. 7
M4	The dividend barrier rises in the asymmetry parameter and in the fixed issuance cost.	$\theta \in \{\lambda_S, K\}$, $F = W$ or N. If $V''(y^*) = 0$, $V'(y^*) = 1$, $\Lambda'(y^*) = 0$, $F'(y^*) = 0$, $\sigma^2(y^*) > 0$, $\rho_L \neq \mu'(y^*)$, $\partial_\theta V = -F$ and the chain rule holds along $y^*(\theta)$, then $dy^*/d\theta = \rho_L F(y^*)/(\rho_L - \mu'(y^*))$, positive when $0 < \rho_L$, $\mu'(y^*) < \rho_L$ and $F(y^*) > 0$.	`Identification.vppp_at_barrier`, `Identification.fpp_at_barrier`, `Identification.barrier_slope`, `Identification.barrier_slope_pos`, `Identification.control_barrier_sign_needs_gap`, `Envelope.wode_of_envelope`, `Envelope.node_of_envelope`, `Envelope.neumann_at_ystar`	p. 8
M5	The recapitalisation trigger rises in the asymmetry parameter and falls in the fixed issuance cost, the injection target rises in the fixed issuance cost, and the trigger-to-target gap widens in it.	If $V''(x)\,dx/d\theta + \partial_\theta V'(x) = 0$ along a boundary $x(\theta)$, $V''(x) \neq 0$ and $\partial_\theta V' = -F'$, then $dx/d\theta = F'(x)/V''(x)$. With $W'(x_L) > 0$, $N'(x_L), N'(y_{\text{post}}) < 0$ and $V''(x_L) > 0 > V''(y_{\text{post}})$, $dx_L/d\lambda_S > 0$, $dx_L/dK < 0$, $dy_{\text{post}}/dK > 0$ and $d(y_{\text{post}} - x_L)/dK > 0$.	`Identification.boundary_slope`, `Identification.trigger_rises_in_lambda`, `Identification.trigger_falls_in_K`, `Identification.target_rises_in_K`, `Identification.gap_increasing_in_K`, `Identification.control_trigger_slope_needs_curvature`, `Envelope.trigger_jump`	p. 8
M6	The Jacobian of the barrier and the trigger in the two parameters is non-singular, because the parameters move the trigger in opposite directions.	$\det \partial(y^*, x_L)/\partial(\lambda_S, K) = \frac{\rho_L}{(\rho_L - \mu'(y^*))V''(x_L)}\big(W(y^*)N'(x_L) - N(y^*)W'(x_L)\big)$, negative under the signs of M4 and M5. For $a_{11}, a_{12}, a_{21} > 0 > a_{22}$, $a_{11}a_{22} - a_{12}a_{21} < 0$. With $a_{22} > 0$ it can vanish.	`Identification.det_formula`, `Identification.det_neg_of_opposite`, `Identification.det_neg`, `Identification.control_same_direction_can_fail`	p. 8

ID	Claim	Statement	Lean	Proof
M7	The value function is convex just above the recapitalisation trigger, strictly so exactly when waiting is strictly suboptimal just below it.	$\Psi = \rho_L V + \Lambda - (1+\kappa)\mu$. If $\sigma^2(x_L) > 0$, $\frac{1}{2}\sigma^2 V'' + \mu V' - \rho_L V - \Lambda = 0$ at x_L^+, $V'(x_L) = 1+\kappa$ and $\mu(1+\kappa) - \rho_L V - \Lambda \le 0$ at x_L, then $\frac{1}{2}\sigma^2 V''(x_L^+) = \Psi(x_L) \ge 0$, and $V''(x_L^+) > 0 \iff \Psi(x_L) > 0$. With $\Psi(x_L) = 0$, $V''(x_L^+) = 0$.	`SecondOrder.psi_identity`, `SecondOrder.psi_nonneg_of_qvi`, `SecondOrder.trigger_convex`, `SecondOrder.trigger_strict_iff`, `SecondOrder.control_trigger_flat_when_psi_zero`	p. 8
M8	The log coordinate maps the state space onto the real line, with the requirement at zero and the distress boundary at minus infinity.	$q > 0 \Rightarrow z$ is strictly increasing on $(1,\infty)$, $1 + qe^{z(x)} = x$ for $x > 1$, $z(1+qe^w) = w$, $z(1+q) = 0$, $z(x) \to -\infty$ as $x \to 1^+$ and $z(x) \to \infty$ as $x \to \infty$.	`CapGeometry.zmap_inverse`, `CapGeometry.zmap_of_inverse`, `Coordinate.zmap_strictMonoOn`, `Coordinate.zmap_at_requirement`, `Coordinate.zmap_tendsto_atBot`, `Coordinate.zmap_tendsto_atTop`	p. 9
M9	Along the cap the drift of the state at the distress boundary is $r - r_L$, and when the drift there is positive the drift of z diverges to plus infinity at that boundary.	$a_1, a_3 > 0$, $a_2 < 1$, $\mu_L = \gamma + r_L \Rightarrow B(1) = r - r_L$. If $B(x) \to b_1 > 0$ as $x \to 1^+$ then $B(x)/(x-1) \to \infty$. With $B \equiv -1$ it tends to $-\infty$.	`Coordinate.cappedDrift_at_one`, `Coordinate.zdrift_tendsto_atTop`, `Coordinate.control_zdrift_needs_positive_limit`, `BCVW.piBar_at_one`, `BCVW.piBar_mul_y`	p. 9
M10	The value function has no kink at the recapitalisation trigger.	If $V \ge \mathscr{M}V$, $V(x_L) = \mathscr{M}V(x_L)$, $\mathscr{M}V$ is affine with slope $1+\kappa$, V has right derivative $V'(x_L^+)$, $\sigma^2(x_L) > 0$, and $V'(x_L^+) > 1+\kappa$ would make $\frac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \le 0$ hold at x_L for every q, then $V'(x_L^+) = 1+\kappa$. If $\frac{1}{2}\sigma^2 q + (1+\kappa)\mu - \rho_L V - \Lambda \le 0$ is needed only for $q \le 0$, it holds when $(1+\kappa)\mu \le \rho_L V + \Lambda$. Without $V \ge \mathscr{M}V$ a downward kink is possible.	`SmoothFit.hasDeriv_Mv`, `SmoothFit.kink_is_convex`, `SmoothFit.no_convex_kink`, `SmoothFit.smooth_fit`, `SmoothFit.control_kink_needs_obstacle`, `Envelope.smf_permits`	p. 9
M11	At $\lambda_S = 1$ the shortfall penalty vanishes identically, and for every $\lambda_S \ge 1$ the running payoff is concave.	$\Lambda(x) = (\lambda_S - 1)(R - x)^+ \Rightarrow \Lambda \equiv 0$ at $\lambda_S = 1$, and $-\Lambda$ is concave on $\mathbb{R}$ for $\lambda_S \ge 1$. At $\lambda_S = 0$, $R = 0$ it is not concave.	`QVI_Part1.lambda_asymmetry_vanishes_at_one`, `QVI_Part1.neg_Lambda_concave`, `QVI_Part1.control_concavity_needs_lambda_ge_one`	p. 9
M12	Balanced growth of equity and requirement pins the return on equity, and the deficit persistence of the maintained model tends to one as growth vanishes, while the persistence asymmetry tends to the surplus payout rate.	$b \ne 0$, $1 + g \ne 0 \Rightarrow$ $\big((1+r_E b)/(1+g) = 1 \iff r_E = g/b\big)$. $\phi^- = 1 - g(1-b)/b \to 1$ as $g \to 0^+$, strictly decreasing in g for $0 < b < 1$. $\phi^- - \phi^+ = \delta_p + g - g/b \to \delta_p$.	`RateBased.balanced_growth_iff`, `RateBased.phi_minus_tendsto_one`, `RateBased.phi_minus_strictAnti`, `RateBased.persistence_asymmetry_eq`, `RateBased.persistence_asymmetry_tendsto`, `RateBased.control_identity_needs_retention`, `RateBased.control_persistence_needs_partial_retention`	p. 10

ID	Claim	Statement	Lean	Proof
M13	The expected time below the requirement falls in the recovery drift and diverges as that drift vanishes.	$d > 0 \Rightarrow m \mapsto d/m$ is strictly decreasing on $(0,\infty)$ and $d/m \to \infty$ as $m \to 0^+$. At $d = 0$ it is constant.	`RateBased.sojourn_strictAnti`, `RateBased.sojourn_tendsto_atTop`, `RateBased.control_sojourn_needs_positive_depth`	p. 10
M14	With a positive fixed issuance cost and a finite value, the discounted number of recapitalisations is finite, so the impulse terms of the verification argument converge absolutely.	$K > 0$, $e_i \geq 0$, $K\sum_{i<n} e_i \leq B$ for all $n \Rightarrow$ $\sum_{i<n} e_i \leq B/K$, $N = \sum_i e_i$ converges and $N \leq B/K$. If also $\lvert c_i\rvert \leq Ce_i$, $\sum_i c_i$ converges absolutely. With $K = 0$, $e_i = 1$ the cost bound holds and N diverges.	`ImpulseCount.partial_sums_bounded`, `ImpulseCount.count_summable`, `ImpulseCount.count_le`, `ImpulseCount.impulse_sum_summable`, `ImpulseCount.control_count_needs_fixed_cost`	p. 10
M15	For every exposure policy that complies with the cap, binding or not, the z-volatility is at most the larger of σ_L^2 and v_1^2 everywhere, and deep in surplus at most the larger of σ_L^2 and v_3^2 up to any margin.	$q > 0$, $a_1 > 0$, $x > 1$, $0 \leq \pi(x)$, $\pi(x)x \leq u(x) \Rightarrow$ $\zeta(x) \leq \max\{\sigma_L^2, v_1^2\}$. If $a_3 > 0$, $a_2 < 1$ and compliance holds for all large x, then for every $\varepsilon > 0$, $\zeta(x) \leq \max\{\sigma_L^2, v_3^2\} + \varepsilon$ for all large x. Both fail with $\pi < 0$ or with $\pi x > u(x)$.	`Bounds.qv_le_max`, `Bounds.zvol2_eq_qv`, `Bounds.nu2_eq_qv`, `Bounds.qv_zero`, `Bounds.deficit_bound`, `Bounds.surplus_bound`, `Bounds.control_bound_needs_long_exposure`, `Bounds.control_bound_needs_compliance`	p. 10

3 Literature

Table 3: Papers read.

ID	Paper	What we rely on	Quote	Page	Rows
L1	Bayraktar et al. [2026]	The cap u, the diffusion $\sigma^2(x,\pi)$ and the crossing point $\bar{x}$ of M2 are this paper's model, its eq. (4) and its switching point, 1.168 at its baseline (p. 15). M2 and M3 are statements about its cap. The parallel ends there, since the paper evaluates no limit of the volatility and has no asymmetry parameter.	"the parameter a3 scales the entire liquidity constraint and determines its asymptotic upper bound"	17	M2, M3, C1, C2
L2	Barberis et al. [2001]	With a constant level of loss aversion, the volatility of log returns equals that of dividend growth for every level (12 percent at four values of b_0, their Table XIII), the parallel to the silence of λ_S in C2. The parallel ends at the mechanism, a constant price-dividend ratio in an endowment economy rather than a binding regulatory cap.	"the volatility of log returns in this model is equal to the volatility of log dividend growth, namely 12 percent."	47	C2

ID	Paper	What we rely on	Quote	Page	Rows
L3	Engle and Siriwardane [2018]	They split observed equity volatility asymmetry into a mechanical-leverage share (3% by their correlation measure) and a market-exposure share, separating a structural source from a non-structural one as C2 does. The parallel ends at the object, since their asymmetry is a response of variance to the sign of news and not a ratio of variances across regions of a state variable.	"Volatility asymmetry is mostly explained by exposure to the aggregate market, not a mechanical leverage effect."	449	C2
L4	Li et al. [2023]	In an unconstrained loss-averse consumption and investment problem, the large-wealth limits of the optimal controls depend on the preference parameters, the contrast to the silence of λ_S in C2. The parallel ends at the constraint, since their risky position is unconstrained and the limit is a Merton-type scaling, while in C2 a binding regulatory cap fixes it. By their own formulas the loss-aversion degree enters those limits only when the curvatures on gains and losses are equal.	"our asymptotic limits differ significantly from the ones in the Merton's problem, which now sensitively depends on the reference degree parameter"	13	C2
L5	Bayraktar et al. [2026]	The paper has no fixed issuance cost, fixes its recapitalisation threshold exogenously, and names a fixed cost as a change that could move recapitalisation. M5 and C4 take up that change. The parallel ends there, since the paper derives no threshold response to a fixed cost and no identification.	"If one introduced fixed issuance costs, endogenous market access, or adverse-selection costs that rise near distress, earlier recapitalization or inaction at the boundary could become optimal."	13	M5, C4
L6	Bayraktar et al. [2026]	A fixed issuance cost charged in dollars breaks the reduction to one dimension. M4 to M6 charge K per unit of liabilities, which keeps it (Appendix D).	"Fixed issuance costs, nonlinear issuance costs, taxes, deposit insurance premia, endogenous deposit rates, balance-sheet size effects, multiple asset classes, or regulatory constraints depending on dollar levels rather than ratios would generally break the one-dimensional reduction."	9	M4, M5, M6

Every paper the tables name has been read.

4 Conclusions

Table 4 holds the conclusions that rest on model rows and on papers read in full. Table 5 holds every claim that a result is new and every other claim still waiting, and each of its rows names what it waits on.

Table 4: Conclusions.

ID	Statement	Rows
C1	Wherever the optimal policy binds the regulatory cap near both ends of the state space, the saturation limits of the z-volatility are v_1^2 and v_3^2. Their ratio involves $(\sigma, \sigma_L, a_1, a_3, c)$ only, so read as the variance ratio λ_V^4 it carries no information about λ_S.	M1, M2, M3, L1
C3	Under the hypotheses listed in Appendix B, the asymmetry parameter λ_S and the fixed issuance cost K are locally identified from the dividend barrier and the recapitalisation trigger. Both parameters raise the barrier, and they move the trigger in opposite directions, so the Jacobian of the two thresholds is non-singular.	M4, M5, M6, M7
C4	The joint identification of C3 is new against the papers read and the literature search of 10 October 2026 recorded in `TODO.md`, whose leads were excluded at the level of their abstracts. Bayraktar et al. [2026] have no fixed issuance cost and set the recapitalisation threshold exogenously (L5). The responses of the barrier and of the target to the fixed issuance cost alone are not claimed new.	M6, L5
C2	That the ratio v_1^2/v_3^2 of the two saturation limits under the two-branch regulatory cap carries no information about the asymmetry parameter is new. The constancy of v_1^2 on the solvency branch and the log coordinate are not claimed new. Among the papers read, Bayraktar et al. [2026] define the cap without evaluating these limits (L1), Barberis et al. [2001] find the level of loss aversion silent in return volatility through a constant price-dividend ratio rather than a binding cap (L2), Engle and Siriwardane [2018] assign observed volatility asymmetry to market exposure rather than mechanical leverage, for a news response rather than a ratio of variances (L3), and Li et al. [2023] find preference parameters in the large-wealth limits of an unconstrained loss-averse control rather than a capped one (L4). The searches of 10 October 2026 recorded in `TODO.md` found no paper that could contain the result, their leads excluded at the level of their abstracts.	M3, L1, L2, L3, L4

Table 5: Pending conclusions.

ID	Statement	Rows

References

Nicholas Barberis, Ming Huang, and Tano Santos. Prospect theory and asset prices. *The Quarterly Journal of Economics*, 116(1):1–53, 2001. doi: 10.1162/003355301556310.

Erhan Bayraktar, Etienne Chevalier, Vathana Ly Vath, and Yuqiong Wang. Tractable bank capital structure: Optimal control under Basel III constraints, 2026. arXiv:2603.14557v2.

Robert F. Engle and Emil N. Siriwardane. Structural GARCH: The volatility-leverage connection. *The Review of Financial Studies*, 31(2):449–492, 2018. doi: 10.1093/rfs/hhx099.

Xun Li, Xiang Yu, and Qinyi Zhang. Optimal consumption with loss aversion and reference to past spending maximum, 2023. arXiv:2108.02648v4.

A Proofs

Notation. In discrete time $V(z)$ is the conditional variance of the shock at log buffer ratio z, with induced variance asymmetry $\lambda_V > 0$, saturation scale $\theta > 0$ and level $V_0 > 0$. In continuous time $x \in (1, \infty)$ is the ratio of assets to liabilities, $q > 0$ the ratio requirement, and $z(x) = \log\big((x-1)/q\big)$. The risky exposure is $\pi(x)$, and the regulatory cap gives the capped exposure $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with crossing point $\bar{x} = (a_3 - a_1 a_2)/(a_3 - a_1)$. The diffusion of x is $\sigma^2(x,\pi) = \pi^2\sigma^2 x^2 + 2\pi c\sigma\sigma_L x(1-x) + \sigma_L^2(1-x)^2$, with c the correlation between the risky return and the liability shock. The z-volatility is $\zeta(x) = z'(x)^2\,\sigma^2\big(x, \pi(x)\big)$, the exposure ratio is $\omega(x) = \pi(x)x/(x-1)$, and $v_i^2 = \sigma^2/a_i^2 - 2c\sigma\sigma_L/a_i + \sigma_L^2$ for $i \in \{1,3\}$. For M4 to M6, V is the value function, y^* the dividend barrier, x_L the recapitalisation trigger and y_{post} the injection target. W is the expected discounted shortfall and N the discounted injection count under the optimal policy, θ is λ_S or K, and F is W or N accordingly. ρ_L is the liability-net discount rate, μ' the derivative in x of the drift along the cap, and $\ell = (R-x)^+$ for $F = W$ and $\ell = 0$ for $F = N$. Primes

are derivatives in x. $B(x) = u(x)(\mu_s - r) + (r - \mu_L)x + \gamma$ is the drift of x along the cap, with $\mu_L = \gamma + r_L$. κ is the proportional issuance cost, $\mathscr{M}V$ the impulse operator, affine with slope $1+\kappa$ on the intervention side, and $\mu = \mu(x_L)$ in M7 and M10. In discrete time, g is the growth rate of required capital, b the retention ratio, r_E the return on equity, ϕ^- and ϕ^+ the deficit and surplus persistence, δ_p the surplus payout rate, and d and m the depth below the requirement and the recovery drift.

Proof of M1

Lean: `RateBased.Vresp_ratio_eq`, `RateBased.tendsto_tanh_atTop`, `RateBased.saturated_ratio_tendsto`, `RateBased.saturated_ratio_lambda4`.

For every z,

$$\frac{V(-z)}{V(z)} = \exp\big(4\log\lambda_V \tanh(z/\theta)\big).$$

$$\tanh y = 1 - \frac{2}{e^{2y}+1} \to 1 \quad (y \to \infty), \qquad z/\theta \to \infty \quad (z \to \infty,\ \theta > 0).$$

$$\exp\big(4\log\lambda_V \tanh(z/\theta)\big) \to \exp(4\log\lambda_V) = \lambda_V^4.$$

θ enters only through $z/\theta \to \infty$, so the limit is free of θ.

□

Proof of M2

Lean: `CapGeometry.hasDerivAt_zmap`, `CapGeometry.zvol2_eq`, `CapGeometry.xbar_sub_one`, `CapGeometry.one_lt_xbar`, `CapGeometry.solvency_binds`, `CapGeometry.liquidity_binds`, `CapGeometry.zvol2_solvency`.

$$\frac{x-1}{a_1} \le \frac{x-a_2}{a_3} \iff (a_3 - a_1)x \le a_3 - a_1 a_2 \iff x \le \bar{x}, \qquad \bar{x} - 1 = \frac{a_1(1-a_2)}{a_3 - a_1} > 0.$$

$$z'(x) = \frac{1}{x-1}, \qquad \zeta(x) = \omega(x)^2\sigma^2 - 2\,\omega(x)\,c\,\sigma\sigma_L + \sigma_L^2.$$

$$1 < x \le \bar{x},\ \pi(x)x = u(x) \ \Rightarrow\ \omega(x) = \frac{1}{a_1} \ \Rightarrow\ \zeta(x) = v_1^2.$$

□

Proof of M3

Lean: `CapGeometry.deficit_limit`, `CapGeometry.surplus_limit`, `CapGeometry.variance_ratio_is_cap_geometry`, `CapGeometry.control_surplus_limit_needs_a1_lt_a3`, `CapGeometry.control_constancy_needs_binding`.

Near 1^+, $\bar{x} > 1$ by M2, so $1 < x < \bar{x}$ eventually and $\zeta(x) = v_1^2$ there. For $x \ge \bar{x}$,

$$\omega(x) = \frac{x-a_2}{a_3(x-1)} = \frac{1}{a_3} + \frac{1-a_2}{a_3}(x-1)^{-1} \to \frac{1}{a_3},$$

$$\zeta(x) = \omega(x)^2\sigma^2 - 2\,\omega(x)\,c\,\sigma\sigma_L + \sigma_L^2 \to v_3^2.$$

Controls, at $\sigma = 1$, $\sigma_L = c = 0$, $q = 1$.

$$a_1 = \tfrac{1}{2},\ a_2 = 0,\ a_3 = \tfrac{1}{4} \ \Rightarrow\ u(x) = 2(x-1),\ \zeta \equiv 4 \ne 16 = v_3^2.$$

$$\pi \equiv 0 \ \Rightarrow\ \zeta(2) = 0 \ne 4 = v_1^2.$$

□

Proof of M4

Lean: `Identification.vppp_at_barrier`, `Identification.fpp_at_barrier`, `Identification.barrier_slope`, `Identification.barrier_slope_pos`, `Identification.control_barrier_sign_needs_gap`, `Envelope.wode_of_envelope`, `Envelope.node_of_envelope`, `Envelope.neumann_at_ystar`.

$$\tfrac{1}{2}\sigma^2 V''' + \tfrac{1}{2}(\sigma^2)' V'' + \mu V'' + \mu' V' - \rho_L V' - \Lambda' = 0, \quad V'' = 0,\ V' = 1,\ \Lambda' = 0 \ \Rightarrow\ \tfrac{1}{2}\sigma^2 V''' = \rho_L - \mu'.$$

$$\tfrac{1}{2}\sigma^2 F'' + \mu F' - \rho_L F + \ell = 0, \quad F'(y^*) = 0,\ \ell(y^*) = 0 \ \Rightarrow\ \tfrac{1}{2}\sigma^2 F'' = \rho_L F.$$

$$V''' \frac{dy^*}{d\theta} + \partial_\theta V'' = 0, \quad \partial_\theta V'' = -F'' \ \Rightarrow\ \frac{dy^*}{d\theta} = \frac{F''}{V'''} = \frac{\rho_L F(y^*)}{\rho_L - \mu'(y^*)}.$$

Control, $\rho_L = 1$, $\mu' = 2$, $F = 1 \Rightarrow dy^*/d\theta = -1$.

□

Proof of M5

Lean: `Identification.boundary_slope`, `Identification.trigger_rises_in_lambda`, `Identification.trigger_falls_in_K`, `Identification.target_rises_in_K`, `Identification.gap_increasing_in_K`, `Identification.control_trigger_slope_needs_curvature`, `Envelope.trigger_jump`.

$$V''(x)\frac{dx}{d\theta} + \partial_\theta V'(x) = 0, \quad \partial_\theta V' = -F' \ \Rightarrow\ \frac{dx}{d\theta} = \frac{F'(x)}{V''(x)}.$$

$$\frac{dx_L}{d\lambda_S} = \frac{W'(x_L)}{V''(x_L)} > 0, \qquad \frac{dx_L}{dK} = \frac{N'(x_L)}{V''(x_L)} < 0, \qquad \frac{dy_{\text{post}}}{dK} = \frac{N'(y_{\text{post}})}{V''(y_{\text{post}})} > 0.$$

$$\frac{d(y_{\text{post}} - x_L)}{dK} > 0, \qquad N(x_L) = N(y_{\text{post}}) + 1.$$

Control, $V''(x) = 0$, $F'(x) = 0 \Rightarrow dx = 0$ and $dx = 1$ both solve the chain rule.

□

Proof of M6

Lean: `Identification.det_formula`, `Identification.det_neg_of_opposite`, `Identification.det_neg`, `Identification.control_same_direction_can_fail`.

$$\det\begin{pmatrix} \frac{\rho_L W(y^*)}{\rho_L - \mu'} & \frac{\rho_L N(y^*)}{\rho_L - \mu'} \\ \frac{W'(x_L)}{V''(x_L)} & \frac{N'(x_L)}{V''(x_L)} \end{pmatrix} = \frac{\rho_L}{(\rho_L - \mu')V''(x_L)}\big(W(y^*)N'(x_L) - N(y^*)W'(x_L)\big).$$

$$a_{11}, a_{12}, a_{21} > 0 > a_{22} \ \Rightarrow\ a_{11}a_{22} < 0 < a_{12}a_{21} \ \Rightarrow\ a_{11}a_{22} - a_{12}a_{21} < 0.$$

Control, $a_{11} = a_{12} = a_{21} = a_{22} = 1 \Rightarrow \det = 0$.

□

Proof of M7

Lean: `SecondOrder.psi_identity`, `SecondOrder.psi_nonneg_of_qvi`, `SecondOrder.trigger_convex`, `SecondOrder.trigger_strict_iff`, `SecondOrder.control_trigger_flat_when_psi_zero`.

$$\tfrac{1}{2}\sigma^2 V'' + \mu V' - \rho_L V - \Lambda = 0, \quad V' = 1 + \kappa \ \Rightarrow\ \tfrac{1}{2}\sigma^2 V''(x_L^+) = \rho_L V + \Lambda - (1+\kappa)\mu = \Psi(x_L).$$

$$\mu(1+\kappa) - \rho_L V - \Lambda \le 0 \ \Rightarrow\ \Psi(x_L) \ge 0 \ \Rightarrow\ V''(x_L^+) \ge 0, \qquad \sigma^2 > 0 \ \Rightarrow\ \big(V''(x_L^+) > 0 \iff \Psi(x_L) > 0\big).$$

Control, $\sigma^2 = 1$, $V' = 1$, $\kappa = 0$, $V = \Lambda = \mu = 0$, $\rho_L = 1 \Rightarrow \Psi = 0$, $V''(x_L^+) = 0$.

□

Proof of M8

Lean: `CapGeometry.zmap_inverse`, `CapGeometry.zmap_of_inverse`, `Coordinate.zmap_strictMonoOn`, `Coordinate.zmap_at_requirement`, `Coordinate.zmap_tendsto_atBot`, `Coordinate.zmap_tendsto_atTop`.

$$z(x)=\log\frac{x-1}{q},\qquad 1+qe^{z(x)}=x\ \ (x>1),\qquad z(1+qe^{w})=w.$$

$$1<x<y\ \Rightarrow\ \frac{x-1}{q}<\frac{y-1}{q}\ \Rightarrow\ z(x)<z(y),\qquad z(1+q)=\log 1=0.$$

$$\frac{x-1}{q}\to 0^{+}\ (x\to 1^{+})\ \Rightarrow\ z\to-\infty,\qquad \frac{x-1}{q}\to\infty\ (x\to\infty)\ \Rightarrow\ z\to\infty.$$

□

Proof of M9

Lean: `Coordinate.cappedDrift_at_one`, `Coordinate.zdrift_tendsto_atTop`, `Coordinate.control_zdrift_needs_positive_limit`, `BCVW.piBar_at_one`, `BCVW.piBar_mul_y`.

$$u(1)=\bar\pi(1)\cdot 1=0\ \Rightarrow\ B(1)=(r-\mu_L)+\gamma=r-r_L\qquad(\mu_L=\gamma+r_L).$$

$$B(x)\to b_1>0,\quad (x-1)^{-1}\to\infty\ \ (x\to 1^{+})\ \Rightarrow\ \frac{B(x)}{x-1}\to\infty.$$

Control, $B\equiv -1\Rightarrow B(x)/(x-1)\to-\infty$.

□

Proof of M10

Lean: `SmoothFit.hasDeriv_Mv`, `SmoothFit.kink_is_convex`, `SmoothFit.no_convex_kink`, `SmoothFit.smooth_fit`, `SmoothFit.control_kink_needs_obstacle`, `Envelope.smf_permits`.

$$V\ge \mathscr{M}V,\ \ V(x_L)=\mathscr{M}V(x_L),\ \ (\mathscr{M}V)'=1+\kappa\ \Rightarrow\ V'(x_L^{+})\ge 1+\kappa.$$

$$V'(x_L^{+})>1+\kappa\ \Rightarrow\ \tfrac12\sigma^2 q+(1+\kappa)\mu-\rho_L V-\Lambda\le 0\ \ \forall q,\qquad q=\frac{2(\rho_L V+\Lambda-(1+\kappa)\mu+1)}{\sigma^2}\ \Rightarrow\ 1\le 0.$$

$$\Rightarrow\ V'(x_L^{+})=1+\kappa;\qquad q\le 0,\ \ (1+\kappa)\mu\le\rho_L V+\Lambda\ \Rightarrow\ \tfrac12\sigma^2 q+(1+\kappa)\mu-\rho_L V-\Lambda\le 0.$$

Control, $v(x)=x$, $\kappa=1$, $\mathscr{M}v(x)=2x\Rightarrow v(0)=\mathscr{M}v(0)$, $v'(0^{+})=1<2$.

□

Proof of M11

Lean: `QVI_Part1.lambda_asymmetry_vanishes_at_one`, `QVI_Part1.neg_Lambda_concave`, `QVI_Part1.control_concavity_needs_lambda_ge_one`.

$$\Lambda(x)\big|_{\lambda_S=1}=(1-1)(R-x)^{+}=0.$$

$$-\Lambda(x)=(\lambda_S-1)\big(\min\{x,R\}-R\big),\qquad \lambda_S-1\ge 0,\ \ x\mapsto\min\{x,R\}\ \text{concave}.$$

Control, $\lambda_S=0$, $R=0\Rightarrow -\Lambda(x)=\max\{-x,0\}$, $-\Lambda(0)=0<\frac12=\frac12(-\Lambda(-1))+\frac12(-\Lambda(1))$.

□

Proof of M12

Lean: `RateBased.balanced_growth_iff`, `RateBased.phi_minus_tendsto_one`, `RateBased.phi_minus_strictAnti`, `RateBased.persistence_asymmetry_eq`, `RateBased.persistence_asymmetry_tendsto`, `RateBased.control_identity_needs_retention`, `RateBased.control_persistence_needs_partial_retention`.

$$\frac{1+r_E b}{1+g} = 1 \iff r_E b = g \iff r_E = \frac{g}{b}.$$

$$\phi^-(g) = 1 - \frac{g(1-b)}{b} \to 1 \ \ (g \to 0^+), \qquad g_1 < g_2,\ 0 < b < 1 \ \Rightarrow\ \phi^-(g_1) > \phi^-(g_2).$$

$$\phi^- - \phi^+ = 1 - \frac{g(1-b)}{b} - (1-\delta_p) = \delta_p + g - \frac{g}{b} \to \delta_p.$$

Controls, $b = 0$, $g = 1 \Rightarrow$ no r_E; $b = 1 \Rightarrow \phi^- \equiv 1$.

□

Proof of M13

Lean: `RateBased.sojourn_strictAnti`, `RateBased.sojourn_tendsto_atTop`, `RateBased.control_sojourn_needs_positive_depth`.

$$0 < m_1 < m_2 \ \Rightarrow\ \frac{d}{m_1} > \frac{d}{m_2}, \qquad \frac{d}{m} = d\,m^{-1} \to \infty \ \ (m \to 0^+).$$

Control, $d = 0 \Rightarrow d/m \equiv 0$.

□

Proof of M14

Lean: `ImpulseCount.partial_sums_bounded`, `ImpulseCount.count_summable`, `ImpulseCount.count_le`, `ImpulseCount.impulse_sum_summable`, `ImpulseCount.control_count_needs_fixed_cost`.

$$K\sum_{i<n} e_i \le B,\ K > 0 \ \Rightarrow\ \sum_{i<n} e_i \le \frac{B}{K} \ \ \forall n, \qquad e_i \ge 0 \ \Rightarrow\ N = \sum_i e_i \le \frac{B}{K} < \infty.$$

$$|c_i| \le C e_i \ \Rightarrow\ \sum_i |c_i| \le CN < \infty.$$

Control, $K = 0$, $e_i = 1 \Rightarrow 0 \le 0$ for every n and $\sum_i e_i = \infty$.

□

Proof of M15

Lean: `Bounds.qv_le_max`, `Bounds.zvol2_eq_qv`, `Bounds.nu2_eq_qv`, `Bounds.qv_zero`, `Bounds.deficit_bound`, `Bounds.surplus_bound`, `Bounds.control_bound_needs_long_exposure`, `Bounds.control_bound_needs_compliance`.

$$\zeta(x) = Q(\omega), \qquad Q(\omega) = \omega^2\sigma^2 - 2\omega c\sigma\sigma_L + \sigma_L^2, \qquad \omega = \frac{\pi(x)\,x}{x-1},$$

$$Q(0) = \sigma_L^2, \qquad Q(1/a_i) = v_i^2.$$

$$\omega = tM,\ t \in [0,1] \ \Rightarrow\ Q(\omega) = (1-t)Q(0) + tQ(M) + \sigma^2 M^2(t^2 - t) \le \max\{Q(0), Q(M)\}.$$

$$0 \le \pi, \ \ \pi x \le u(x) \le \frac{x-1}{a_1} \ \Rightarrow\ 0 \le \omega \le \frac{1}{a_1} \ \Rightarrow\ \zeta(x) \le \max\{\sigma_L^2, v_1^2\}.$$

$$\pi x \le \frac{x-a_2}{a_3} \ \Rightarrow\ \omega \le w(x) = \frac{1}{a_3} + \frac{1-a_2}{a_3(x-1)} \to \frac{1}{a_3}, \qquad Q(w(x)) \to v_3^2.$$

Controls, $(\sigma,\sigma_L,c)=(1,1,1)$, $\omega=-1\Rightarrow Q=4>1=\max\{Q(0),Q(1)\}$; $(\sigma,\sigma_L,c)=(1,0,0)$, $\omega=2>1\Rightarrow Q=4>1$.

□

B Evidence and verification

Toolchain. Lean `v4.32.0-rc1` with Mathlib at tag `v4.32.0-rc1`, commit `360da6fa66c1`, pinned in `lean/mathlib/lake-manifest.json`. Every file under `lean/mathlib/` is a build root. A theorem passes the axiom audit when it depends on no axiom outside `propext`, `Classical.choice` and `Quot.sound`.

Checks. `verify.sh` runs `checks/verify_manuscript.py`, `checks/verify_mathlib.py` and `checks/verify_docs.py`, and writes their output to `VERIFICATION.md`. `lit/verify_lit.sh` runs the suite of every literature record.

What Lean does not carry. Two inputs enter the theorems as hypotheses or definitions. The z-volatility ζ is defined as $z'(x)^2\sigma^2(x,\pi(x))$, and that this is the diffusion coefficient of $z(X_t)$ is Itô's formula, which is not proved (M2, M3). That the optimal policy binds the cap near both ends of the state space is a hypothesis of M3, and C1 states it as its condition. `PROOFS_v2.tex` supports it only by the numerical solve, which is evidence and not proof.

M4 to M6 take as hypotheses, in raw form, the inputs that `PROOFS_v2.tex` establishes analytically or assumes. That V is the value function is the verification theorem, not proved. Differentiability of V and of the thresholds in θ is the regularity lemma and the implicit function theorem. The envelope identities $\partial_{\lambda_S}V=-W$ and $\partial_K V=-N$ are Danskin's theorem. The interior equation enters differentiated in x and in θ. The signs $W(y^*)>0$, $N(y^*)>0$, $N'<0$ on the inaction region and $W'(x_L)>0$ come from hitting-time arguments, and $W'(x_L)>0$ holds exactly when $\rho_L W(y_{\text{post}})<R-x_L$. $V''(x_L)>0$ follows from M7 given $\Psi(x_L)>0$, the strict form of the inequality the quasi-variational inequality gives on the intervention side. $V''(y_{\text{post}})<0$ is a hypothesis, since the derivation of it in `PROOFS_v2.tex` fails (Appendix E). $\rho_L>\mu'(y^*)$ is a parameter condition. C3 states these as its hypotheses.

M9 takes the drift of z to be $B(x)/(x-1)-\nu_1^2/2$, its Itô form, and does not classify the boundary $x=1$, so whether it is reached is not proved. M10 takes the viscosity supersolution property of V at x_L as a hypothesis, in the form that a kink admits lower test functions of every curvature. M13 takes the expected first-passage time of a random walk with drift m from depth d to be d/m, Wald's identity, which is not proved. M14 takes the cost bound as a hypothesis, that each impulse costs at least K and the strategy's value is finite, with $e_i=e^{-\rho_L\tau_i}$ the discounted impulse times and c_i the impulse terms of the verification argument. Interior regularity of V, C^2 on the inaction region and C^3 away from R and $\bar{x}$, is assumed wherever a row differentiates V; that $\sigma^2(x,\pi)>0$ for $x\neq 1$ and $c^2<1$ is `BCVW.sig2_nondegenerate`. The lower bound $V(x)\geq x-1$ of `PROOFS_v2.tex` (Prop. LQB) is used by no row and is not carried.

Illustrations. No row rests on SymPy or on numerical sampling.

C Terminology

Term	Symbol	Meaning
Induced variance asymmetry	λ_V	The asymmetry read off the variance ratio, distinct from λ_S.
Variance ratio	λ_V^4	The ratio of conditional variances deep in deficit to deep in surplus (M1).
Saturation scale	θ	How sharply the conditional variance moves between its deficit and surplus levels.
Asymmetry parameter	λ_S	The weight on shortfall below the reference in the objective.
Regulatory cap	$u(x)$	The capped exposure $\min\{(x-1)/a_1,(x-a_2)/a_3\}$.
Cap geometry	a_1,a_2,a_3	The parameters of the regulatory cap.
Solvency and liquidity regime	$x\leq\bar{x}$, $x\geq\bar{x}$	The ranges where the first and the second branch of u bind (M2).
z-volatility	$\zeta(x)$	The squared diffusion coefficient of z (Appendix B).
Saturation limits	ν_1^2,ν_3^2	The limits of ζ at the distress boundary and deep in surplus (M3).

Term	Symbol	Meaning
Fixed issuance cost	K	The cost of an issue that does not scale with its size, per unit of liabilities (Appendix D).
Dividend barrier	y^*	The state at which dividends are paid.
Recapitalisation trigger	x_L	The state at which capital is raised.
Injection target	y_{post}	The state reached by an issue.
Expected discounted shortfall	W	$\mathbb{E}_x \int_0^\infty e^{-\rho_L t}(R - X_t)^+ dt$ under the optimal policy, $-\partial_{\lambda_s} V$.
Discounted injection count	N	$\mathbb{E}_x \sum_i e^{-\rho_L \tau_i}$ under the optimal policy, $-\partial_K V$.
Proportional issuance cost	κ	The cost of an issue per unit raised.
Impulse operator	$\mathscr{M}V$	The value of issuing at once to the best target, net of both issuance costs.

D Primitives and scope

The regulatory cap. The capped exposure is $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with $0 < a_1 < a_3$ and $a_2 < 1$, taken from `PROOFS_v2.tex`. Under these restrictions the solvency branch binds near the distress boundary and the liquidity branch deep in surplus (M2). With $a_1 \geq a_3$ the surplus limit of M3 fails, by its control. Rows M2, M3 and C1 depend on it.

The variance ratio in continuous time. `PROOFS_v2.tex` reads the discrete-time λ_V^4 of M1 in the continuous-time model as v_1^2/v_3^2, identifying the conditional variance $V(z)$ deep in deficit and deep in surplus with the two saturation limits of ζ. This is a definition, not a result, and no theorem connects the discrete-time recursion to the continuous-time diffusion. C1 depends on it.

The persistence coefficients. $\phi^- = 1 - g(1-b)/b$ and $\phi^+ = 1 - \delta_p$ are the maintained model of `PROOFS_v2.tex`, a proportional control calibrated to the accounting identity, and are not derived from full retention. M12 depends on them.

The fixed issuance cost. K is charged per unit of liabilities, so the dollar cost of an issue is K times liabilities. This keeps the reduction of the bank's problem to the single state x, which a fixed cost in dollars would break (L6). M4 to M6, C3 and C4 depend on it.

E Refuted conjectures

R1, strict concavity at the injection target. `PROOFS_v2.tex` (Prop. SOC(i)) derives $V''(y_{\text{post}}) < 0$ from the sign change of $\varphi' = V' - (1+\kappa)$ at y_{post}, positive below and negative above, and from $\varphi \in C^3$, arguing that $\varphi''(y_{\text{post}}) = 0$ would give a zero of even order. A zero of odd order can still have $\varphi'' = 0$. With $\varphi(y) = -y^4/4$ and $y_{\text{post}} = 0$, $\varphi'(y) = -y^3$ is positive for $y < 0$ and negative for $y > 0$, while $\varphi''(0) = 0$ (`SecondOrder.refuted_target_strict`). The weak inequality survives and strictness does not, so $V''(y_{\text{post}}) < 0$ enters M5 as a hypothesis (Appendix B). C3 does not use it.

R2, a vanishing persistence asymmetry. `PROOFS_v2.tex` (Prop. PSA) states that the persistence asymmetry $\phi^- - \phi^+$ vanishes with g. With $\phi^+ = 1 - \delta_p$ it equals $\delta_p + g - g/b$ and tends to δ_p (M12), so it vanishes only when $\delta_p = 0$. At $b = \delta_p = \frac{1}{2}$ the limit is $\frac{1}{2}$, not 0 (`RateBased.refuted_asymmetry_vanishes`). M12 states the limit.